

AI Usage Note: Quantitative Research & Event-Driven Backtesting

Candidate: Quant Engineer Intern Applicant
Submission to: AlgoChowk
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1. Tools Used

- **Antigravity AI (powered by Google DeepMind's advanced reasoning models):** Used as an interactive quant pair-programmer for scaffolding Python code modules, formulating statistical tests, generating vector arithmetic, and drafting LaTeX/Markdown formulas.
- **Python Data Science Ecosystem:** pandas, numpy, scipy.stats, matplotlib, seaborn for empirical execution.

2. How the AI Tools Were Used

1. **Code Scaffolding:** Generating modular Python boilerplate (src/data_loader.py, src/event_engine.py, src/statistical_tests.py, src/backtester.py) following clean software engineering design patterns.
2. **Vectorized Computation:** Accelerating forward-return indexing and implementing the 5,000-iteration bootstrap resampling loop using vectorized NumPy array slicing.
3. **Statistical Formulations:** Rapidly writing standard implementations for Welch's two-sample t -test, Mann-Whitney U rank tests, and Bonferroni p -value adjustments.
4. **Data Visualization:** Drafting publication-grade multi-panel Matplotlib and Seaborn visualization code with confidence intervals and dual-axis log equity curves.

3. Human Decisions (Where I Guided the AI)

- **Falsification-First Mindset:** AI chatbots instinctively try to "make the strategy profitable" by fine-tuning thresholds or looking for complex indicators to generate high returns. I explicitly instructed the research framework to adopt an **unbiased, scientific falsification approach**—prioritizing truth and empirical reality over artificial backtest optimization.
- **Enforcing Observation Independence:** I identified that consecutive crash days (e.g., during the 2008 Lehman collapse or March 2020) violate the independent and identically distributed (i.i.d.) assumption of standard hypothesis tests. I directed the creation of a configurable **lockout window** to evaluate non-overlapping events.
- **Unconditional Baseline Benchmarking:** AI initially evaluated post-event returns simply against zero ($\$H_0: \mu = 0$). I overrode this, pointing out that because equity indices possess positive natural drift ($+0.044\%/day$ in NIFTY), testing against zero is misleading. I forced all comparative tests to evaluate excess return against the **unconditional market baseline** over the exact same holding horizons.

4. Suggestions I Changed or Disagreed With

- **Threshold Overfitting:** The AI initially suggested scanning thresholds in 0.1% increments to find the "optimal" threshold. I rejected this because post-hoc parameter optimization causes severe **data snooping bias** (P-hacking). Instead, I constrained the analysis to pre-registered, economically intuitive thresholds (-1.0%, -1.5%, -2.0%, -2.5%, -3.0%).
- **Ignoring Overnight Gapping:** The AI assumed holding returns could simply be computed from $\$T$ Close to $\$T+h$ Close. I pointed out that entering at $\$T$ Close creates look-ahead bias unless a Market-On-Close (MOC) mechanism is explicitly modeled. I required a dual-model comparison ($\$T$ Close vs. $\$T+1$ Open) to expose the opening gap slippage.

5. Incorrect AI Suggestions Identified

1. **Look-Ahead Execution Bias:** An early script draft generated by the AI executed orders on day $\$T$ using day $\$T$'s Open price—an egregious temporal leak (buying at morning open based on an event confirmed only at afternoon close). I caught and corrected this immediately, ensuring signals at $\$T$ Close are strictly executed at $\$T+1$ Open.

2. **Exclusion of Statutory Indian Transaction Costs:** Standard AI backtesting templates omitted Indian market statutory friction (Securities Transaction Tax / STT, exchange turnover fees, stamp duty). I implemented an explicit 10 bps round-trip friction and 5 bps per leg slippage model, which revealed that nominal gross profits were completely wiped out.
 3. **Overlapping Position Compounding:** The AI's initial backtesting loop accumulated leverage by stacking positions whenever multiple drops occurred back-to-back, artificially amplifying returns during recoveries and blowing up the account during crashes. I replaced this with an explicit single-position state-machine backtester.
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6. Key Learnings & Takeaways

- **AI is an Accelerator, Not a Scientist:** AI is remarkably effective at generating bug-free vectorized syntax and statistical scaffolding, saving hours of manual coding. However, it lacks domain-grounded skepticism. Without strict human domain oversight, AI will readily optimize flawed strategies, introduce subtle look-ahead biases, and mistake statistical noise for alpha.
- **The Value of Falsification:** The most impactful quantitative research often lies in disproving intuitive market dogmas ("always buy the dip"). Rigorous falsification prevents deploying capital into structurally flawed strategies.